

University of Texas at Austin, Jackson School of Geosciences
GEO 325C/398C, 2026
Continuum Mechanics

Midterm Exam – ANSWERS [21 pts]

- (1) [10 pts] **The one with Force (Balance):** Isostasy can explain the formation of basins through extension and thinning of continental crust as shown in Figure 1. Variables with subscript c denote the normal crust and variables with subscript b denote the thinned crust in the basin. The variable h is the total thickness of the crust, z is the elevation the crust rises above the mantle and b the depth the crust is submerged in the mantle. The initial width of the thinned crust is w_0 and the final width of the basin is w . The depth of the basin is $\Delta z = z_c - z_b$. The densities ρ_c and ρ_m of crust and mantle are constant.

- (a) Use area/mass conservation of the area shown in gray to obtain an expression for h_b [2 pts].

The initial gray area must equal the final gray area, so that $w_0 h_c = w h_b$. Hence the relation between stretched and unstretched crust is

$$h_b = \frac{w_0}{w} h_c.$$

- (b) Use the hydrostatic force balance on unstretched crust to obtain an expression for b_c . The gravitational body force, i.e., weight, is given by $f_G = m_c \mathbf{g}$, where m_c is the mass of the crust. The hydrostatic surface force, i.e., buoyancy, is given by $f_B = -m_d \mathbf{g}$, where m_d is the mass of the displaced mantle [2 pts].

The hydrostatic force balance is $\mathbf{f}_G = -\mathbf{f}_B$. The mass of the original crust is $m_c = A h_c \rho_c$ and the mass of the displaced mantle is $m_d = A b_c \rho_m$. Substituting into the force balance we obtain

$$A h_c \rho_c \mathbf{g} = A b_c \rho_m \mathbf{g}.$$

Solving for h_b we obtain

$$b_c = \frac{\rho_c}{\rho_m} h_c.$$

- (c) Use the hydrostatic force balance on the stretched crust in the basin to obtain an expression for b_b and use the results in (a) to express it in terms of h_c [2 pts].

The hydrostatic force balance is $\mathbf{f}_G = -\mathbf{f}_B$. The mass of the stretched crust is $m_c = A h_b \rho_c$ and the mass of the displaced mantle is $m_d = A b_b \rho_m$. Substituting into the force balance we obtain

$$A h_b \rho_c \mathbf{g} = A b_b \rho_m \mathbf{g}.$$

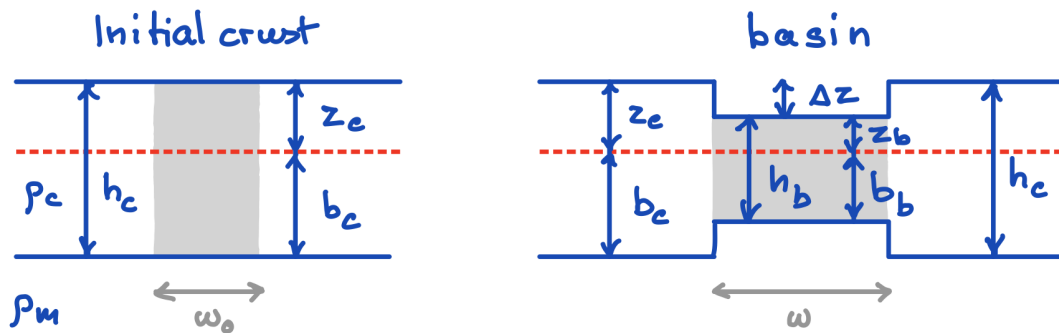


FIGURE 1. Geometry of the isostasy problem for basin formation. The gray area is being stretched and thinned.

Solving for h_b we obtain

$$b_b = \frac{\rho_c}{\rho_m} h_b = \frac{\rho_c}{\rho_m} \frac{w_0}{w} h_b.$$

- (d) Use the variables to derived above to find an expression for the depth of the basin Δz in terms of h_c , ρ_c/ρ_m and w_0/w [**3 pts**].

$$\begin{aligned}\Delta z &= z_c - z_b, \\ &= (h_c - b_c) - (h_b - b_b), \\ &= \left(1 - \frac{\rho_c}{\rho_m}\right) h_c - \left(\frac{w_0}{w} - \frac{\rho_c}{\rho_m} \frac{w_0}{w}\right) h_c, \\ &= \left(1 - \frac{\rho_c}{\rho_m}\right) \left(1 - \frac{w_0}{w}\right) h_c\end{aligned}$$

- (e) Use the following rounded parameters $\rho_c/\rho_m = 0.9$, $w_0/w = 0.5$ and $h_c = 30$ km to compute Δz using the formula derived above [**1 pt**].

With these numbers the depth of the basin is

$$\Delta z = (1 - 0.9) \cdot 0.5 \cdot 30 \text{ km} = 1.5 \text{ km}.$$

- (2) [**6 pts**] **The One with Rotations:** A general rotation around axis, $\mathbf{r}$, by the clockwise angle, θ , is given by

$$\mathbf{Q}(\hat{\mathbf{r}}, \theta) = \hat{\mathbf{r}} \otimes \hat{\mathbf{r}} + \cos \theta (\mathbf{I} - \hat{\mathbf{r}} \otimes \hat{\mathbf{r}}) + \sin \theta \mathbf{R}(\hat{\mathbf{r}}),$$

where $\mathbf{R}(\hat{\mathbf{r}})$ is the axial tensor

$$\mathbf{R}(\hat{\mathbf{r}}) = \epsilon_{ijk} r_k \mathbf{e}_i \otimes \mathbf{e}_j = \begin{bmatrix} 0 & -\hat{r}_3 & \hat{r}_2 \\ \hat{r}_3 & 0 & -\hat{r}_1 \\ -\hat{r}_2 & \hat{r}_1 & 0 \end{bmatrix}.$$

Answer the questions below for a general rotation, i.e., do not assume a particular axis or angle and do **not** use index notation for this problem.

- (a) Show that the axis of rotation, $\hat{\mathbf{r}}$, remains unchanged by rotation, $\mathbf{Q}\hat{\mathbf{r}} = \hat{\mathbf{r}}$. Use the dyadic property, $(\mathbf{a} \otimes \mathbf{b})\mathbf{c} = (\mathbf{b} \cdot \mathbf{c})\mathbf{a}$ and definition of the axial tensor, $\mathbf{R}(\hat{\mathbf{r}})\mathbf{u} = \hat{\mathbf{r}} \times \mathbf{u}$ [**3 pts**].

From the definition of the rotation tensor we have

$$\begin{aligned}\mathbf{Q}(\hat{\mathbf{r}}, \theta) &= [\hat{\mathbf{r}} \otimes \hat{\mathbf{r}} + \cos \theta (\mathbf{I} - \hat{\mathbf{r}} \otimes \hat{\mathbf{r}}) + \sin \theta \mathbf{R}(\hat{\mathbf{r}})] \hat{\mathbf{r}}, \\ &= (\hat{\mathbf{r}} \otimes \hat{\mathbf{r}})\hat{\mathbf{r}} + \cos \theta (\mathbf{I}\hat{\mathbf{r}} - (\hat{\mathbf{r}} \otimes \hat{\mathbf{r}})\hat{\mathbf{r}}) + \sin \theta \mathbf{R}(\hat{\mathbf{r}})\hat{\mathbf{r}}, \\ &= (\hat{\mathbf{r}} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} + \cos \theta (\hat{\mathbf{r}} - (\hat{\mathbf{r}} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}}) + \sin \theta (\hat{\mathbf{r}} \times \hat{\mathbf{r}}), \\ &= \hat{\mathbf{r}},\end{aligned}$$

where the last equality comes from $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = 1$ and $\hat{\mathbf{r}} \times \hat{\mathbf{r}} = \mathbf{0}$.

- (b) Show that the transpose of a rotation tensor, $\mathbf{Q}^T(\hat{\mathbf{r}}, \theta)$, is identical to the rotation tensor with the negative angle, $\mathbf{Q}(\hat{\mathbf{r}}, -\theta)$, i.e., $\mathbf{Q}^T(\hat{\mathbf{r}}, \theta) = \mathbf{Q}(\hat{\mathbf{r}}, -\theta)$. Use the transpose of the dyadic product, $(\mathbf{a} \otimes \mathbf{b})^T = \mathbf{b} \otimes \mathbf{a}$, and the skew symmetry of the axial tensor, $\mathbf{R}(\hat{\mathbf{r}})^T = -\mathbf{R}(\hat{\mathbf{r}})$ [**3 pts**].

From the definition of the rotation tensor, the transpose is given by

$$\begin{aligned}\mathbf{Q}^T(\hat{\mathbf{r}}, \theta) &= [\hat{\mathbf{r}} \otimes \hat{\mathbf{r}} + \cos \theta (\mathbf{I} - \hat{\mathbf{r}} \otimes \hat{\mathbf{r}}) + \sin \theta \mathbf{R}(\hat{\mathbf{r}})]^T, \\ &= (\hat{\mathbf{r}} \otimes \hat{\mathbf{r}})^T + \cos \theta (\mathbf{I}^T - (\hat{\mathbf{r}} \otimes \hat{\mathbf{r}})^T) + \sin \theta \mathbf{R}(\hat{\mathbf{r}})^T, \\ &= (\hat{\mathbf{r}} \otimes \hat{\mathbf{r}}) + \cos \theta (\mathbf{I} - \hat{\mathbf{r}} \otimes \hat{\mathbf{r}}) - \sin \theta \mathbf{R}(\hat{\mathbf{r}}).\end{aligned}$$

From the definition of the rotation tensor, the tensor with the negative angle is given by

$$\begin{aligned}\mathbf{Q}(\hat{\mathbf{r}}, -\theta) &= \hat{\mathbf{r}} \otimes \hat{\mathbf{r}} + \cos \theta (\mathbf{I} - \hat{\mathbf{r}} \otimes \hat{\mathbf{r}}) + \sin \theta \mathbf{R}(\hat{\mathbf{r}}), \\ &= \hat{\mathbf{r}} \otimes \hat{\mathbf{r}} + \cos(-\theta) (\mathbf{I} - \hat{\mathbf{r}} \otimes \hat{\mathbf{r}}) + \sin(-\theta) \mathbf{R}(\hat{\mathbf{r}}), \\ &= \hat{\mathbf{r}} \otimes \hat{\mathbf{r}} + \cos(\theta) (\mathbf{I} - \hat{\mathbf{r}} \otimes \hat{\mathbf{r}}) \sin(\theta) \mathbf{R}(\hat{\mathbf{r}}),\end{aligned}$$

where the last line results from $\cos(-\theta) = \cos(\theta)$ and $\sin(-\theta) = -\sin(\theta)$. Comparing the two terms we see that $\mathbf{Q}^T(\hat{\mathbf{r}}, \theta) = \mathbf{Q}(\hat{\mathbf{r}}, -\theta)$.

(3) [5 pts] **The One with Index Notation:** As always, follow the basic pattern:

- Define the vectors, in terms of coefficients and basis vectors, i.e., $\mathbf{a} = a_i \mathbf{e}_i$.
- Substitute these definitions into the expressions.
- In dot and cross products, pull out the coefficients to the front so that you have products between base vectors ($\mathbf{e}_i$'s).
- Use the definitions of dot and cross products to evaluate the products between the base vectors: $\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$ and $\mathbf{e}_i \times \mathbf{e}_j = \epsilon_{ijk} \mathbf{e}_k$.
- Simplify the expression further and eliminate any δ_{ij} or ϵ_{ijk} .
- Convert back from index to dyadic notation.

Rewrite the expression below to replace the cross products with dot products. Use index notation and at the end convert the answer back to dyadic notation.

$$(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = \dots,$$

where $\mathbf{a} = a_i \mathbf{e}_i$, $\mathbf{b} = b_j \mathbf{e}_j$, $\mathbf{c} = c_p \mathbf{e}_p$ and $\mathbf{d} = d_q \mathbf{e}_q$. The $\delta\epsilon$ -identity is given by $\epsilon_{ijk}\epsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}$.

Substitute and simplify

$$\begin{aligned}(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) &= (a_i \mathbf{e}_i \times b_j \mathbf{e}_j) \cdot (c_p \mathbf{e}_p \times d_q \mathbf{e}_q), \\ &= a_i b_j c_p d_q (\mathbf{e}_i \times \mathbf{e}_j) \cdot (\mathbf{e}_p \times \mathbf{e}_q), \\ &= a_i b_j c_p d_q (\epsilon_{ijk} \mathbf{e}_k) \cdot (\epsilon_{pql} \mathbf{e}_l), \\ &= \epsilon_{ijk} \epsilon_{pql} a_i b_j c_p d_q (\mathbf{e}_k \cdot \mathbf{e}_l), \\ &= \epsilon_{ijk} \epsilon_{pql} a_i b_j c_p d_q \delta_{kl}, \\ &= \epsilon_{ijk} \epsilon_{pqk} a_i b_j c_p d_q, \\ &= (\delta_{ip} \delta_{jq} - \delta_{iq} \delta_{jp}) a_i b_j c_p d_q, \\ &= \delta_{ip} \delta_{jq} a_i b_j c_p d_q - \delta_{iq} \delta_{jp} a_i b_j c_p d_q, \\ &= a_i b_j c_i d_j - a_i b_j c_j d_i, \\ &= a_i c_i b_j d_j - a_i d_i b_j c_j, \\ &= (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c}).\end{aligned}$$